

Solution: Let $A = [a_1 \ a_2 \ \dots \ a_n]$ where each $a_i \in \mathbb{R}^m$.

$$\text{Im } A = \text{Im} [a_1 \ a_2 \ \dots \ a_n] = [\text{Im } a_1 \ \text{Im } a_2 \ \dots \ \text{Im } a_n] \text{ (By defn. of matrix multiplication)} \\ = [a_1 \ a_2 \ \dots \ a_n] \text{ (By defn. of identity matrix Im)} \\ = A \quad \square$$

32) Show that $AIn = A$ where A is an $m \times n$ matrix.

Solution:

34) Give a formula for $(ABx)^T$, where x is a vector and A and B are matrices of appropriate sizes.

Solution: Let A be a $m \times n$ matrix. By defn. of matrix multiplication, B must have n rows. Let B have r columns. By defn. of matrix multiplication, AB is a $m \times r$ matrix. x is a vector, which is itself a column matrix. By defn. of matrix multiplication, x must have r rows.

$$\begin{aligned} (ABx)^T &= ((AB)x)^T = x^T (AB)^T \text{ (Theorem 3(d))} \\ &= x^T (B^T A^T) \text{ (Theorem 3(d))} \\ &= (x^T B^T) A^T \text{ (Theorem 2(a))} \\ &= x^T B^T A^T \end{aligned}$$

$$\therefore (ABx)^T = x^T B^T A^T \quad \square$$

33) Prove Theorem 3(d): $(AB)^T = B^T A^T$

Solution:

Let A be a $m \times n$ matrix. By defn. of matrix multiplication, B must have n rows. Let B have r columns.

By the Row-Column Rule, the (i,j) th entry of (AB) is: $a_{i1} \cdot b_{1j} + a_{i2} \cdot b_{2j} + \dots + a_{in} \cdot b_{nj}$

By the defn. of Transpose, the (i,j) th entry of AB is the (j,i) th entry of $(AB)^T$. $\therefore$ The (j,i) th entry of $(AB)^T$ is $\sum_{k=1}^n a_{ik} \cdot b_{kj}$.

By the Row-Column Rule, the (j,i) th entry of $B^T A^T$ is: $b'_{j1} \cdot a'_{1i} + b'_{j2} \cdot a'_{2i} + \dots + b'_{jn} \cdot a'_{ni} = \sum_{k=1}^n b'_{jk} \cdot a'_{ki}$